

AMENDMENT NO. 2 NOVEMBER 1973
TO
IS : 4697-1968 METHODS OF MEASUREMENTS ON
GEIGER-MÜLLER COUNTER TUBES

Alterations

(*Page 4, clause 2.2*) — Substitute the following for the existing clause
' 2.2 Pulse — A brief change in current or voltage, such as results from the passage of an ionising particle or quanta (or two or more simultaneous ones) through a counter tube.

2.2.1 Spurious Pulse — A pulse generated without the passage of ionising particle or quanta.

(*Page 4, clause 2.3*) — Substitute the following for the existing clause:
' 2.3 Count — The number of pulses recorded in a specified period. '

(*Page 4, clause 2.5, clause heading*) — Substitute '**Background**' for '**Background of a Counter Tube**'.

(*Page 4, clause 2.6*)

a) *Line 1* — Substitute 'any event' for 'an event'.

b) *Line 3* — Delete the words ' (see also 2.2) '.

(*Page 4, clause 2.8.1, line 2*) — Substitute 'process under stated conditions' for 'process'.

(*Page 4, clause 2.9, Note, line 1*) — Substitute 'gas multiplication factor' for 'gas multiplication (amplification) '.

(*Page 5, clause 2.15, line 3*) — Delete the words ' as a constituent of counting gas '.

(*Page 5, clause 2.19, line 3*) — Substitute 'counter' for 'counting'.

(*Page 5, clause 2.20, line 2*) — Substitute 'probability of the occurrence' for 'the occurrence'.

(*Page 6, clauses 2.23 and 2.24*) — Transpose these clauses and re-number ' 2.24 ' as ' 2.23 ' and ' 2.23 ' as ' 2.24 '.

(*Page 6, clause 2.26, line 2*) — Substitute ' in the applied voltage (usually 100 V) ' for ' usually 100 V in the applied voltage '.

[*Page 6, clause 2.33 (re-numbered 2.35), clause heading*] — Substitute the following for the existing heading:

' Needle Counter Tube '

(Page 13, clause 4.4.1)

- a) Line 1 — Substitute 'in' for 'with'.
- b) Lines 4 and 5 — Substitute 'across the counter tube' for 'at the amplifier input'.

(Page 13, clause 4.5, line 2) — Substitute 'in the' for 'with'.

(Page 14, clause 4.6, line 5) — Substitute ' V_a ' for ' V_s '.

(Page 14, clause 4.8)

- a) Heading — Delete the words 'Count Rate'.
 - b) Line 2 — Substitute 'in the' for 'with'.
 - c) Line 4 — Substitute '35 mm' for '3.5 mm'.
- (Page 14, clause 4.9, line 2) — Substitute 'in the' for 'with'.

(Page 15, clause 4.10)

- a) Line 3 — Substitute 'Fig. 7 and 8' for 'Fig. 4'.
- b) Lines 6 and 7 — Substitute 'amplitude' for 'effective charge per counting event'.
- c) Line 7 — Substitute 'change' for 'charge'.

(Page 15, clause 4.11.1)

- a) Line 2 — Substitute 'in' for 'with' and delete the word 'also'.
- b) Line 7 — Substitute 'counter tube' for 'counting tube'.

(Pages 15 and 16, Fig. 9 and 10) — Substitute the following for the existing figures:

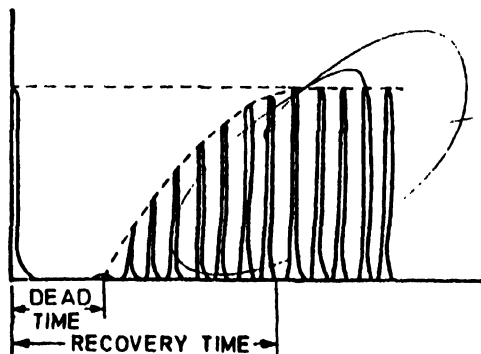


FIG. 9 TYPICAL OSCILLOGRAPH SHOWING DEAD TIME AND
RECOVERY TIME OF ORGANIC-VAPOUR QUENCHED
COUNTER TUBE

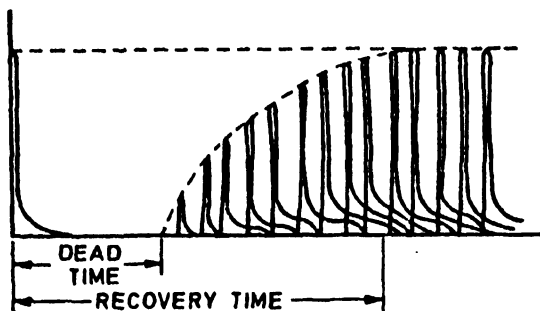


FIG. 10 TYPICAL OSCILLOGRAPH SHOWING DEAD TIME AND RECOVERY TIME OF HALOGEN QUENCHED COUNTER TUBE

(Page 16, clause 4.11.2, second para, line 2) — Delete the word ' also '.

(Pages 16 and 17, clause 4.12.1, fourth para)

a) Line 2 — Delete the words ' period of '.

b) Line 3 — Substitute ' count ' for ' counting '.

c) Note 1, line 2 — Substitute ' exposure-rate ' for ' dose-rate '.

d) Note 1, third para, line 1 — Substitute ' tubes ' for ' tube ' and add the following new paragraph under this note:

' Precautions should be taken to avoid Compton effect due to the counter-tube-source geometry. '

e) Note 2, second para — Substitute ' is variable ' for ' should be made '.

f) Note 3, line 3 — Substitute ' build-up cap should be ' for ' build-up '.

(Page 17, clause 4.12.2, Note) — Substitute the following for the existing note:

Note — The exposure-rate by a point source of gamma radiation is given by:

$$\text{Exposure-rate in mr/h} = \frac{\tau}{d^2} \times \text{activity in millicuries}$$

where τ is a constant for the radioactive material used which is internationally accepted by the authorized organizations and d is the distance in centimetres (for Co^{60} : $\tau = 13.2 \times 10^3$).

(Page 17, clause 4.12.3)

a) Line 2 — Substitute ' exposure-rate ' for ' dose-rate ' and delete the words ' counts/Rontgen or more usually in '.

b) Second para, line 3 — Delete the word ' therefore '.

(Pages 17 and 18, clause 4.13) — Substitute the following for the existing clause:

‘4.13 Temperature Coefficient of Starting Voltage and Plateau Threshold Voltage — If the relationship over the required operating range is non-linear, then it may be necessary to define different coefficients over different temperature ranges, and the starting voltage should be measured at appropriate temperature intervals over the operating temperature range.

The results are plotted graphically and from this graph the variation in temperature coefficient over given temperature ranges can be calculated.

If the temperature coefficient is linear over the operating temperature range, the starting voltage is measured at two different temperatures which are chosen to be near the ends of the operating temperature range.

If the starting voltages are V_{s1} and V_{s2} at temperatures T_1 and T_2 respectively, then the temperature coefficient is given by:

$$\frac{V_{s2} - V_{s1}}{\frac{1}{2}(V_{s2} + V_{s1})} \times \frac{100}{(T_2 - T_1)} \text{ percent per deg C.}$$

Since the difference between the threshold voltage and the starting voltage is approximately constant, the temperature coefficient of the starting voltage will also be the temperature coefficient of the threshold voltage.

(Page 18, clause 4.14) — Substitute the following for the existing clause:

‘4.14 Temperature Coefficient of Radiation Sensitivity — If the relationship over the required operating range is non-linear, then it may be necessary to define different coefficients over temperature ranges, and the sensitivity should be measured at appropriate temperature intervals over the operating temperature range.

The results are plotted graphically and from this graph the variation in temperature coefficient over the given temperature range can be calculated.

If the temperature coefficient is linear over the operating temperature range, the sensitivity is measured at two different temperatures which are chosen to be near the ends of the operating temperature range.

If the sensitivities are S_1 and S_2 at temperatures T_1 and T_2 respectively, then the temperature coefficient is given by:

$$\frac{S_2 - S_1}{(S_2 + S_1)} \times \frac{100}{T_2 - T_1} \text{ percent per deg C.}$$

Addenda

(Page 6, clause 2.27) — Add the following new clause after 2.27 and re-number the subsequent clauses accordingly:

'2.28 Useful Area — Part of the window of a counter tube for which a particle or an incident quanta may be the origin of an output pulse.'

[Page 6, clause 2.29 (re-numbered 2.30), line 2] — Add 'or quanta' after 'particles'.

[Page 6, clause 2.29 (re-numbered 2.30)] — Add the following new clause after re-numbered 2.30 and re-number the subsequent clauses accordingly:

'2.31 Detection Efficiency — Ratio of the number of counts to the total number of particles or of quantas falling on the useful area, when the counting rate is sufficiently low to make a correction of dead time unnecessary.'

(Page 8, clause 3.1.2) — Add the following new clause after 3.1.2:

'3.1.2.1 The discharge of the halogen quenched counter tube is strongly influenced by the impedance of the tube (resistance and capacitance). Moreover, the number of recorded counts is strongly influenced by the measuring equipment (input sensitivity, input impedance, and time constant).

(Page 13, clause 4.2.2) — Add the following note under this clause:

'NOTE — When measuring counter tubes having a dead time greater than 500 μ s (estimated at the recommended working voltage), the radiation field should be reduced so that the count rate is approximately

$$\frac{5 \times 10^4}{(\text{dead time in } \mu\text{s})} \cdot \text{counts per second}'$$

(Page 13, clause 4.4.3) — Add the following note under this clause:

'NOTE — The measurement value may depend upon the sensitivity threshold of the measuring circuit.'

(Page 14, clause 4.9) — Add the following note at the end of the second para:

'NOTE — The counting rate is expressed in counts per second.'